American Computer Science League

2024-2025 ● Contest 1: Short Problem Solutions ● Elementary Division

1. Computer Number Systems 2024_8 = 2*8^3 + 2*8^1 + 4 = 1044_{10}	D. 1044
2. Computer Number Systems Converting from hexadecimal to octal is easiest if you go through base 2. Every hex digit is 4 binary digits and every octal digit is 3 binary digits so regrouping the binary digits is done as follows: $A27_{16} = 1010\ 0010\ 0111_2 = 101\ 000\ 100\ 111_2 = 5047_{16}$	C. 5047
3. Computer Number Systems Converting 198_{10} to base 16 is done by repeatedly dividing by the number 16 and getting the remainder. Reading the remainder from last to first always produces the answer: 198/16 = 12 r 6, 12/16 = 0 r 12 so the answer is C6	E. C6
4. Computer Number Systems Counting in binary from 20_{10} to 24_{10} is: 20 = 10100, 21 = 10101, 22 = 10110, 23 = 10111, 24 = 11000 There are a total of 14 1's.	B. 14
5. Computer Number Systems The two colors $B0E7C3_{16}$ and $98FDAE_{16}$ represent the red, green, and blue components of 2 different colors. The red difference is B0 - 98 = 18_{16} because 16 - 8 = 8 and 10 - 9 = 1. The green difference is FD - E7 = 16_{16} because 13 - 7 = 6 and 15 - 14 = 1. The blue difference is C3 - AE = 15_{16} because 16 + 3 - 14 = 5 and 11 - 10 = 1. Differences determine the varying shade of each component. The largest variation is 18_{16}.	C. 18

6. Computer Number Systems The red component of any hexadecimal color is based on the first 2 digits only. For the Olympic Rings 0081C8 (water blue), 008651 (shamrock green), 000000 (black), FCB131 (macaroni and cheese), and EE334E (dark carmine pink), the first 2 digits are 00 for 3 of them: blue, green, and black.	D. 3